

2. Maxillary second premolar:

Maxillary Second Premolar



2. Maxillary second premolar:

- The maxillary 2nd premolar has two cusps (facial & lingual) of nearly equal size and height
- Shorter mesial cuspal ridges
- This tooth typically has one root with 2 pulp canals.
- The facial ridge is not as prominent as that of the maxillary 1st premolar.
- This tooth will usually present with more supplemental occlusal anatomy.

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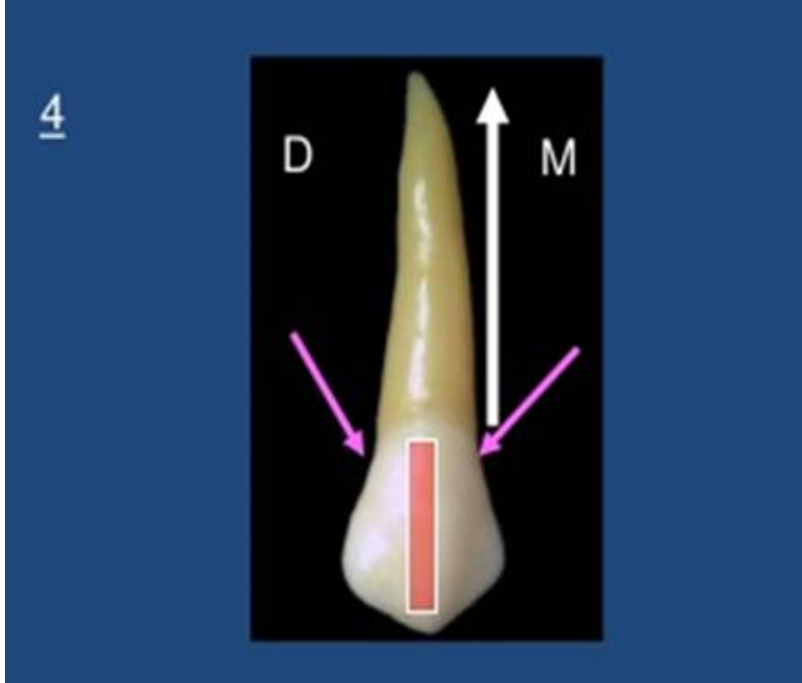
Buccal aspect

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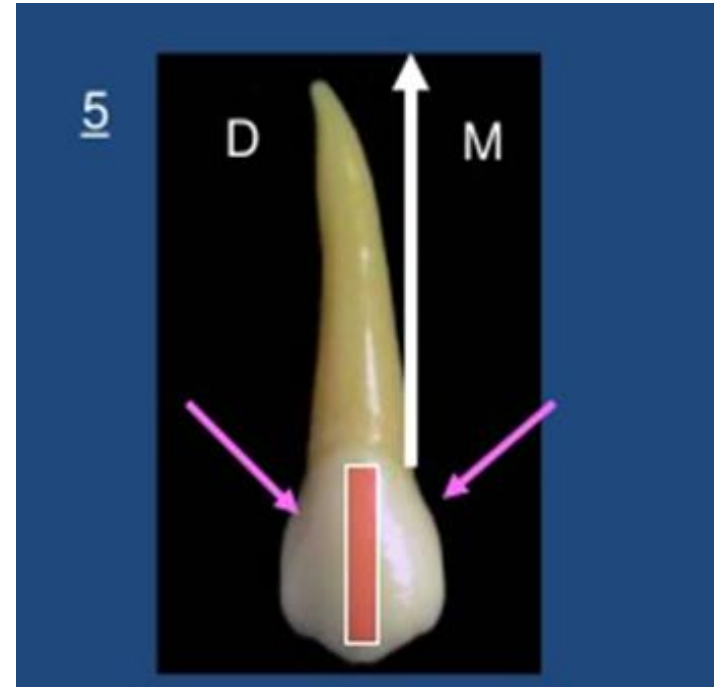


- B cusp is long and pointed.
 - M slope is longer than D
 - Mesial contact A is in the M1/3 while distally more occlusal.
 - Prominent B ridge.
- B cusp is short and less pointed.
 - M slope is shorter than D
 - Mesial contact A is in O1/3 while distally more cervically.
 - Less prominent B ridge.

Buccal aspect



- Narrow cervical portion.
- Short root.



- Wider cervical portion.
- Longer root

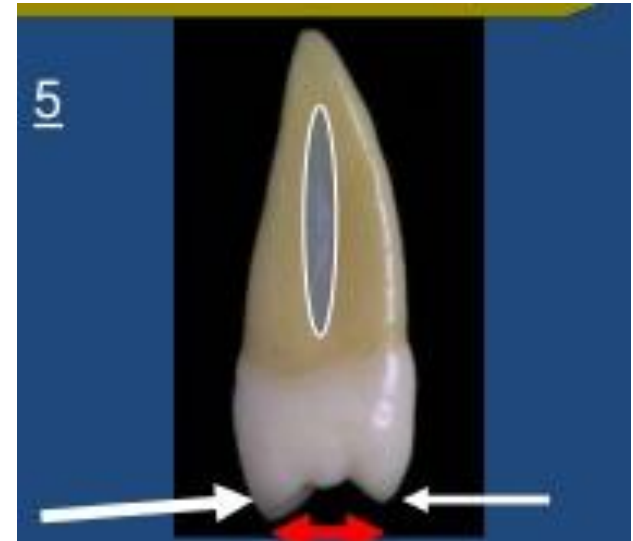
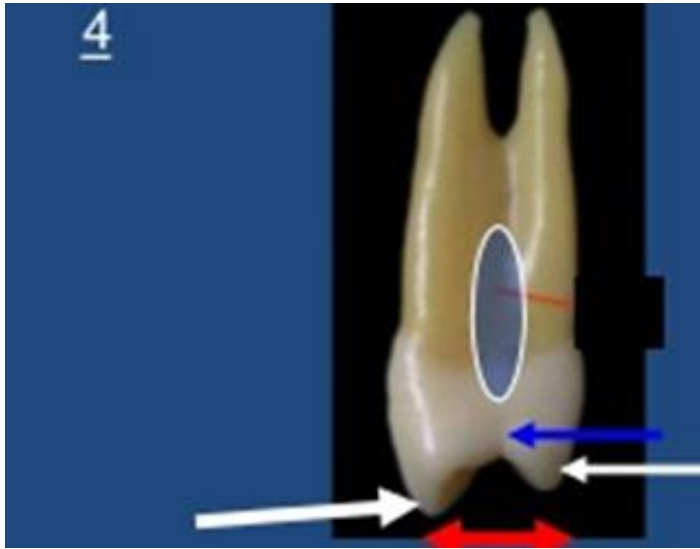
Lingual aspect.



- The L cusp is shorter by 1 mm than the B cusp.
- 80% has 2 roots with L root is shorter than B root and its apex is pointed.

- The L and B cusps are nearly of same height.
- Mostly it has one root.
- Rare to has 2 roots. In case of two roots the L root is shorter and its apex is more blunt.

Mesial aspect



- B cusp is longer than L cusp by 1mm
 - The occlusal table is wide
 - MMRDG and canine fossa.
- The cusps are nearly at the same level.
 - The occlusal table is narrow
 - The crown surface is smooth and convex shallow developmental depression.

Mesial aspect

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- ✓ MMR at the junction of M1/3 & O1/3
- ✓ Contact area: At the middle third (OC dimension) and slightly buccal to the midline (BL dimension).

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- ✓ MMR positioned more occlusal
- ✓ At the occlusal third (OC dimension) and slightly buccal to the midline (BL dimension).

Distal aspect

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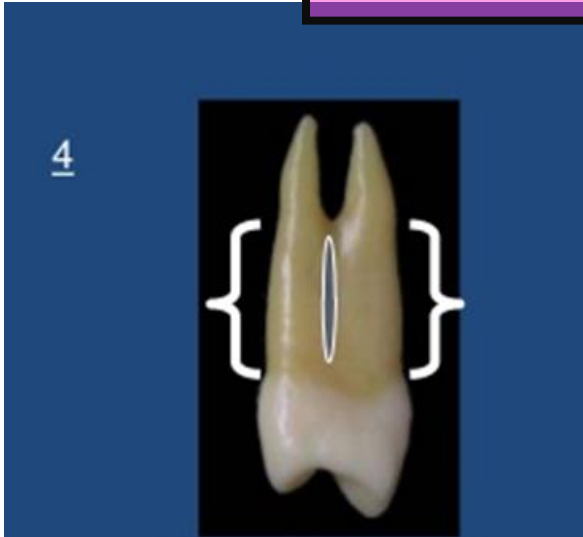
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- ✓ DMR more occlusal than MMR
- ✓ C.A. occlusally positioned and more buccally than MCA.
- ✓ Smooth and convex surface except for a small flat area cervical to contact area

- ✓ DMR more cervical than MMR
- ✓ C.A. cervically positioned and more buccally than MCA.
- ✓ Smooth and convex surface.

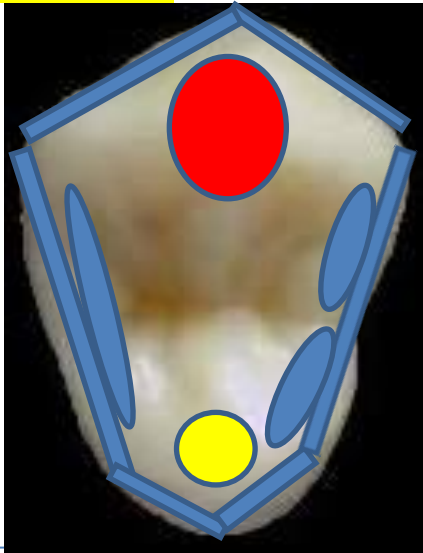
Distal aspect



- Root trunk is long as the bifurcation is near apical 1/3.
 - Surface is smooth and convex except shallower DD on the root trunk than mesially.
- One root and if 2 the bifurcation will be more apically.
 - Surface is smooth and convex except deeper DD in the middle of the root than mesially.

Occlusal aspect

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It's hexagonal

B & L triangular ridges.

M & D marginal ridges

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It's oval

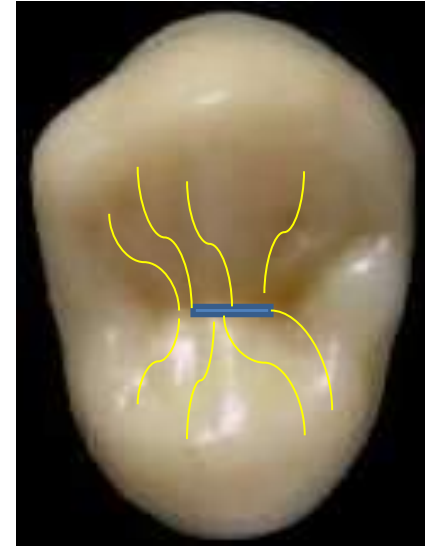
B & L triangular ridges

Thicker M & D MRs

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Occlusal aspect

5



✓ Long Central DG

✓ MMRDG developmental groove

✓ Not present.

✓ Shorter Central DG

✓ Not present

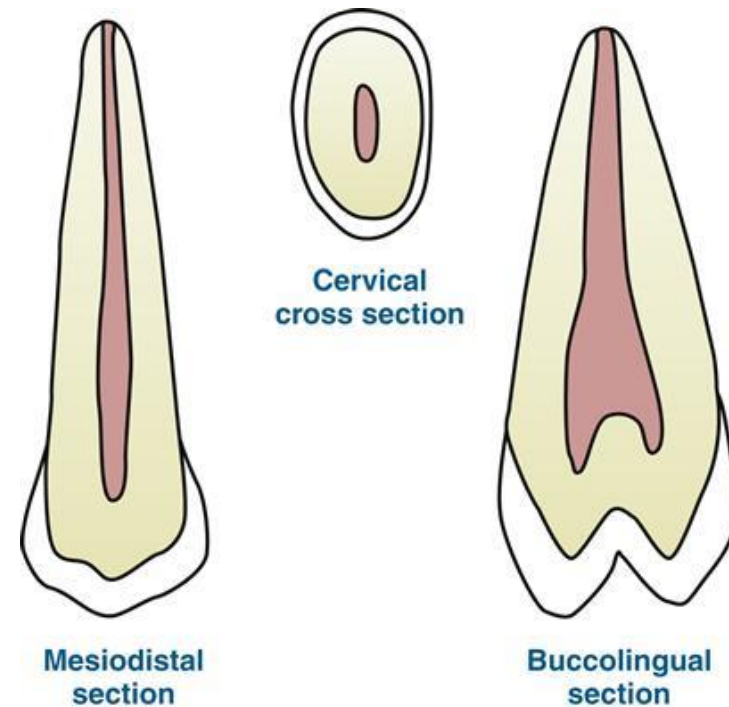
✓ Multiple supplemental grooves.



Pulp cavity

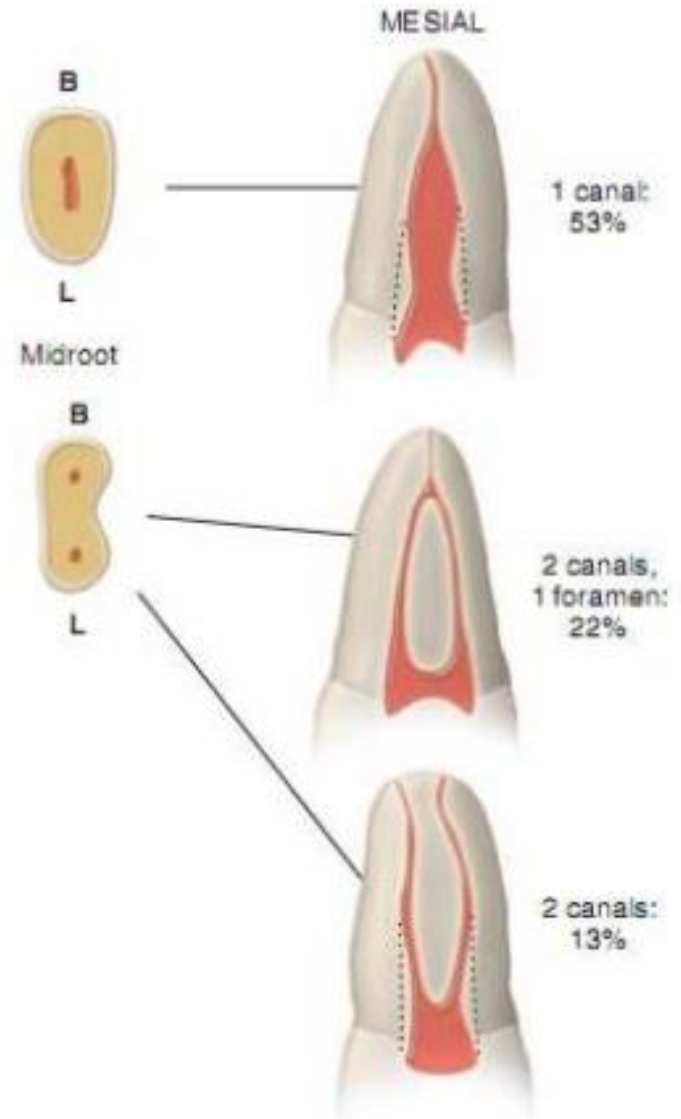
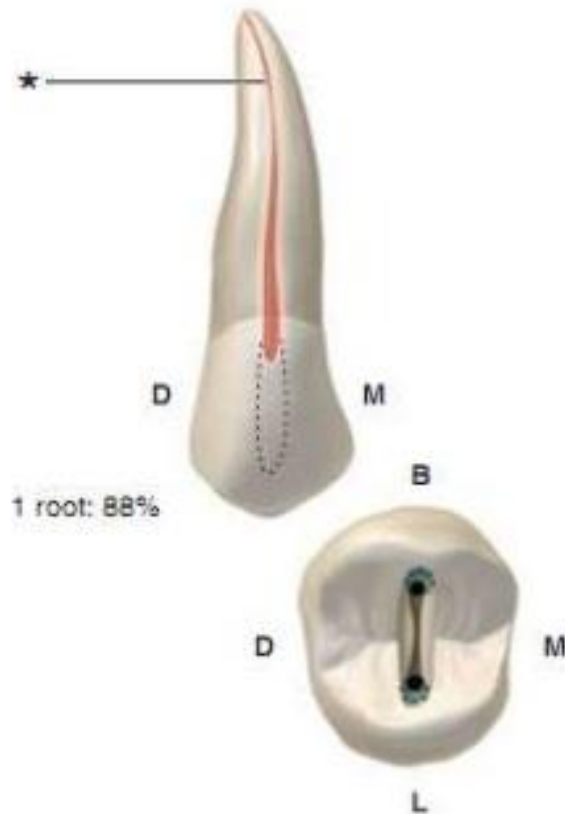
- ✓ The root canal system of the maxillary second premolar is wider buccolingually than mesiodistally.
- ✓ This tooth may have one or two roots and canals
- ✓ The mesiodistal and buccolingual aspects of the pulp chamber are similar to those of the first premolar.
- ✓ Abuccal and a palatal pulp horns are present; the buccal pulp horn is larger.

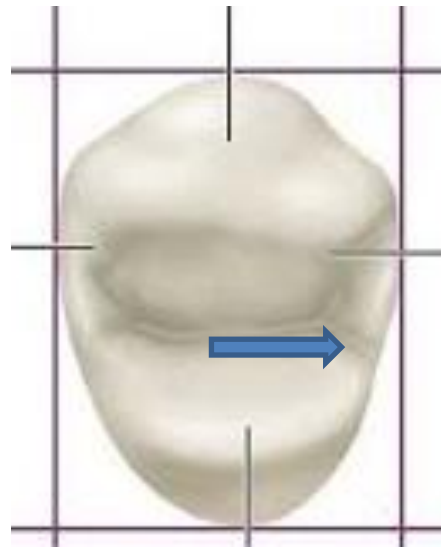
✓ **Cervical cross section:** is oval



Pulp cavity

Maxillary Right Second Premolar





A

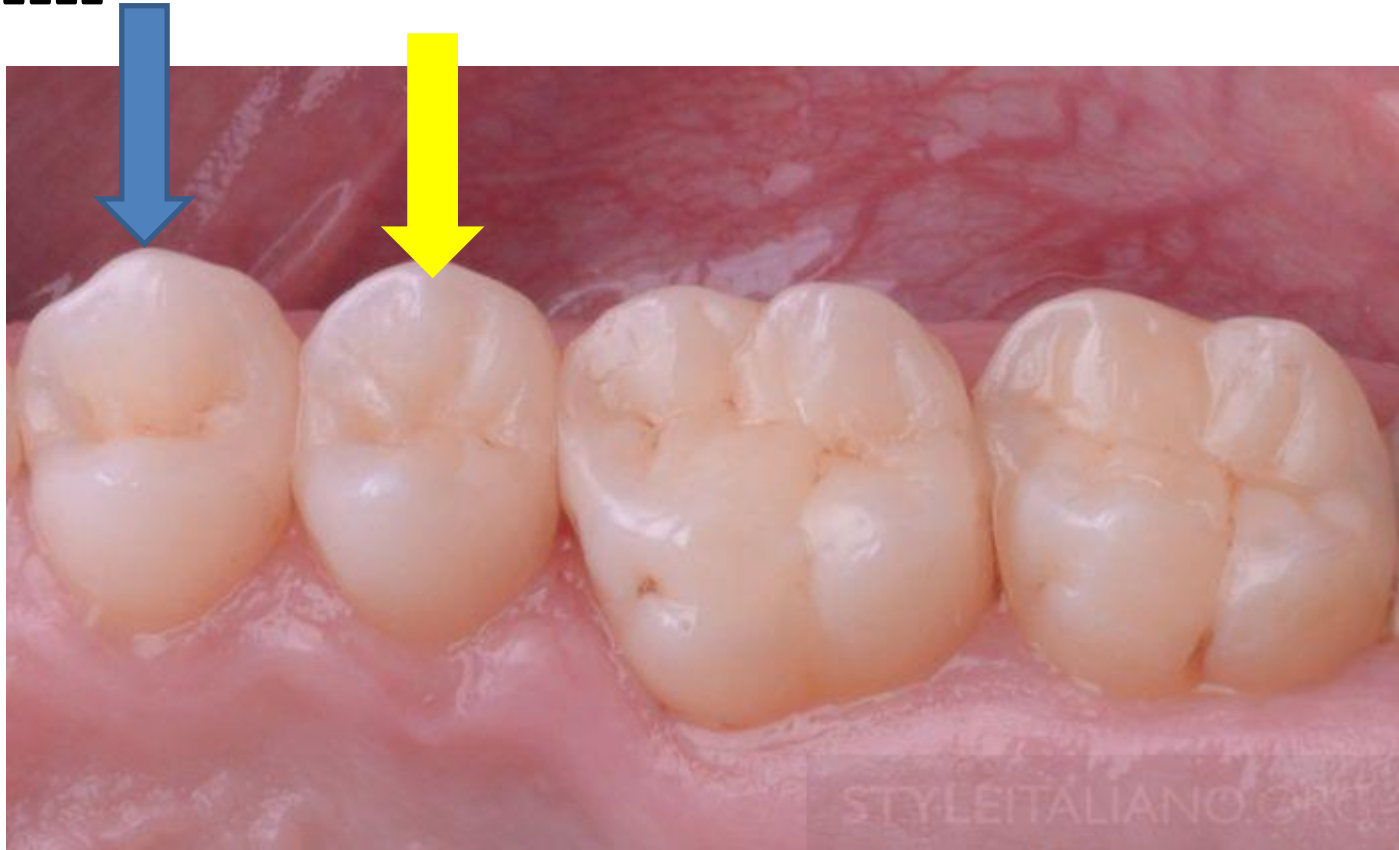


B

- Name A ,B?
- Name the aspect?
- The arrows point at?

✓ The yellow arrow is pointed to -----

✓ The blue arrow is pointed to-----



Thank
you